ISA Design Part 2
Computer Architecture

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ISA of Branching

For Every Jump instructions except JMPREG,

Opcode (6 bit)		Address
2 bits	4 bits	7 bits
Types of instruction	Operations	Value (0000000-1111111)

12 7	6 0
10xxxx (Opcode)	Address

For JMPREG instruction,

Opcode (6 bit)		Register 1	Unused
2 bits	4 bits	3 bits	4 bits
Types of instruction	Operations	Ra (000-111)	XXXX

ISA of Branching

Opcode		Address	Register 1	Assembly Example
Type (2 bits)	Operations (4 bits)	7 bits	3 bits	
10	0000 (JMP)	0000000-1111111 (0-127)	X	JMP LABEL
	0001 (JE)	0000000-1111111 (0-127)	X	JE LABEL
	0010 (JNE)	0000000-1111111 (0-127)	X	JNE LABEL
	0011 (JL)	0000000-1111111 (0-127)	X	JL LABEL
	0100 (JLE)	0000000-1111111 (0-127)	X	JLE LABEL
	0101 (JG)	0000000-1111111 (0-127)	X	JG LABEL
	0110 (JGE)	0000000-1111111 (0-127)	X	JGE LABEL
	0111 (JC)	0000000-1111111 (0-127)	X	JC LABEL
	1000 (JNC)	0000000-1111111 (0-127)	X	JNC LABEL
	1001 (JZ)	0000000-1111111 (0-127)	X	JZ LABEL
	1010 (JNZ)	0000000-1111111 (0-127)	X	JNZ LABEL
	1011 (JMPREG) (It jumps to address saved in R0-R4 register)	X	000-100 (0-4) (Last 4 bits are unused)	JMPREG R1
	1100 (XXX)	X	X	XXX
	1101 (XXX)	X	X	XXX

Example: ISA of Branching

12	11	10	09	08	07	06	05	04	03	02	01	00
10XXXX (Opcode)						Address						

Question:

Translate following assembly code to machine code using ISA:

1. `JMP 3(LABEL)`

Answer:

Since type of instruction is Branching, so opcode of first two bits[12:11] is 10.

12	11	10	09	08	07	06	05	04	03	02	01	00
1	0	X	X	X	X	X	X	X	X	X	X	X

Since opcode of JMP is 0000, so rest of opcode[10:7] is 0000.

12	11	10	09	08	07	06	05	04	03	02	01	00
1	0	0	0	0	0	X	X	X	X	X	X	X

Since address is 3, it will be in binary [6:0] is 0000011.

12	11	10	09	08	07	06	05	04	03	02	01	00
1	0	0	0	0	0	0	0	0	0	0	1	1

Example: ISA of Branching

12	11	10	09	08	07	06	05	04	03	02	01	00
10XXXX (Opcode)						Address						

Question:

Translate following assembly code to machine code using ISA:

2. JLE 100(LABEL)

Answer:

Since type of instruction is Branching, so opcode of first two bits[12:11] is 10.

12	11	10	09	08	07	06	05	04	03	02	01	00
1	0	X	X	X	X	X	X	X	X	X	X	X

Since opcode of JMP is 0100, so rest of opcode[10:7] is 0100.

12	11	10	09	08	07	06	05	04	03	02	01	00
1	0	0	1	0	0	X	X	X	X	X	X	X

Since address is 100, it will be in binary [6:0] is 1100100.

12	11	10	09	08	07	06	05	04	03	02	01	00
1	0	0	1	0	0	1	1	0	0	1	0	0

Memory Instructions
Addressing Modes Examples
Register Indirect Mode/Indexed Mode/Based Mode

Main Memory (RAM)

Memory Address	Memory Contents
000 (0)	000 000 0000 100 (4)
001 (1)	000 000 0000 010 (2)
010 (2)	000 000 0000 101 (5)
011 (3)	000 000 0000 111 (7)
100 (4)	000 000 0000 110 (6)
101 (5)	000 000 0000 011 (3)
110 (6)	000 000 0001 010 (10)

`LOAD R0, [R2]`

Step 1

Step 2

Step 3

Register Set

Register Address	Register Name	Register Contents
000 (0)	R0	0100 (4)
001 (1)	R1	0001 (1)
010 (2)	R2	0000 (0)
011 (3)	R3	0100 (4)
100 (4)	R4	0101 (5)
101 (5)	R5	0010 (2)
110 (6)	R6	0010 (2)

Step 1: R2=0
Step 2: [R2]=[0]=4
Step3: R0 = 4

Memory Instructions
Addressing Modes Examples
Register Indirect Mode/Indexed Mode/Based Mode

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STORE [R3+R0+3], R2
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Main Memory (RAM)

Memory Address	Memory Contents
000 (0)	000 000 0000 100 (4)
001 (1)	000 000 0000 010 (2)
010 (2)	000 000 0000 101 (5)
011 (3)	000 000 0000 111 (7)
100 (4)	000 000 0000 110 (6)
101 (5)	000 000 0000 011 (3)
110 (6)	000 000 0000 000 (0)

Step 1

Step 3

R3+R0+3

Step 2

Register Set

Register Address	Register Name	Register Contents
000 (0)	R0	0001 (1)
001 (1)	R1	0001 (1)
010 (2)	R2	0000 (0)
011 (3)	R3	0010 (2)
100 (4)	R4	0101 (5)
101 (5)	R5	0010 (2)
110 (6)	R6	0010 (2)

Step 1: R3=2,R0=1, R3+R0+3=2+1+3=6

Step 2: [R3+R0+3]=[6]

Step3: [6] = R2 = 0

ISA of Memory Instructions

For Direct Mode (`LOAD R0, [2]/STORE [3], R0`),

Opcode (6 bit)		Register 1	Address/Displacement
2 bits	4 bits	3 bits	4 bits
(11) Types of instruction	Operations	Ra(000-111)	Disp (0000-1111)

For Register Indirect Mode (`LOAD R0, [R2]/STORE [R3], R0`),

Opcode (6 bit)		Register 1	Register 2	Unused
2 bits	4 bits	3 bits	3 bits	1 bits
(11) Types of instruction	Operations	Ra(000-111)	Rb (000-111)	X

For Based with Displacement Mode (`LOAD R0, [R2+2]/STORE [R3+2], R0`),

Opcode (6 bit)		Register 1	Register 2	Displacement
2 bits	4 bits	3 bits	2 bits	2 bits
(11) Types of instruction	Operations	Ra(000-111)	Rb (00-11)	Disp(00-11)

ISA of Memory & IO Instructions

Opcode			Register 1	Register 2	Address	Assembly Example
Type (2 bits)	Operations (4 bits)	Type of Operations	RA	RB	Disp	
11	0000 (LOAD)	Direct Mode	000-111 (0-7)	X	0000-1111 (0-15)	LOAD RA,[Disp]
	0001 (LOAD)	Register Indirect Mode/Indexed Mode/Based Mode	000-111 (0-7)	000-111 (0-7)	X	LOAD RA, [RB]
	0010 (LOAD)	Based/Indexed with Displacement Mode	000-111 (0-7)	00-11 (0-3)	00-11 (0-3)	LOAD RA, [RB+Disp]
	0011 (STORE)	Direct Mode	000-111 (0-7)	X	0000-1111 (0-15)	STORE [Disp], RA
	0100 (STORE)	Register Indirect Mode/Indexed Mode/Based Mode	000-111 (0-7)	000-111 (0-7)	X	STORE [RB], RA
	0101 (STORE)	Based/Indexed with Displacement Mode	000-111 (0-7)	00-11 (0-3)	00-11 (0-3)	STORE [RB+Disp], RA
	1101 (ACCEPT_INPUT)	Waiting for inputs. Send input data to Input Register	X	X	X	ACCEPT_INPUT
	1110 (PRINT_OUTPUT)	Send data to be printed to Output Register	X	X	X	PRINT_OUTPUT
	1111 (PRINT_CLEAR	Clear output display	X	X	X	PRINT_CLEAR

Example: ISA of Memory & IO Instructions

12	11	10	09	08	07	06	05	04	03	02	01	00
01XXXX (Opcode)						Register 1(RA)			Address			

Question:

1. `LOAD R3, [2]` (Format: `LOAD RA,[Disp]`)

Answer:

Since type of instruction is Memory & IO operations, so opcode of first two bits[12:11] is 11.

12	11	10	09	08	07	06	05	04	03	02	01	00
1	1	X	X	X	X	X	X	X	X	X	X	X

Since opcode of this LOAD (Direct mode) operation is 0000, so rest of opcode[10:7] is 0000.

12	11	10	09	08	07	06	05	04	03	02	01	00
1	1	0	0	0	0	X	X	X	X	X	X	X

Since RA register is R3, it will be register 1[6:4] in ISA and its value in binary is 011.

12	11	10	09	08	07	06	05	04	03	02	01	00
1	1	0	0	0	0	0	1	1	X	X	X	X

Since address/displacement is 2, it will be Address[3:0] in ISA and its value in binary is 0010.

12	11	10	09	08	07	06	05	04	03	02	01	00
1	1	0	0	0	0	0	1	1	0	0	1	0

Example: ISA of Memory & IO Instructions

12	11	10	09	08	07	06	05	04	03	02	01	00
01XXXX (Opcode)						Register 1(RA)			Register 2(RB)			Unused

Question:

2. **STORE [R2], R1** (Format: **STORE [RB], RA**)

Answer:

Since type of instruction is Memory & IO operations, so opcode of first two bits[12:11] is 11.

12	11	10	09	08	07	06	05	04	03	02	01	00
1	1	x	x	x	x	x	x	x	x	x	x	x

Since opcode of this STORE (Register Indirect mode) operation is 0100, so rest of opcode[10:7] is 0100.

12	11	10	09	08	07	06	05	04	03	02	01	00
1	1	0	1	0	0	x	x	x	x	x	x	x

Since RA register is R1, it will be register 1[6:4] in ISA and its value in binary is 001.

12	11	10	09	08	07	06	05	04	03	02	01	00
1	1	0	1	0	0	0	0	1	x	x	x	x

Since RB register is R2, it will be register 2[3:1] in ISA and its value in binary is 010.

12	11	10	09	08	07	06	05	04	03	02	01	00
1	1	0	1	0	0	0	0	1	0	1	0	x

Example: ISA Design

Design an ISA of 16-bit CPU based on following requirements:

i. **Types of Instructions:** ALU Instruction (Immediate) (XOR RA, 2), Jump Instruction (JMP LABEL), Memory Instructions (LOAD Direct) (LOAD RA, [Address]).

ii. **No of Registers:** 32

iii. **Supported RAM:** 128x32

Answer:

Word size of RAM is 32. So, maximum size of instruction is 32.

No of types of Instructions is 3 and Assuming no of ALU Instructions (Immediate) and Jump Instructions is at least 16. So, Opcode is 6 bit where 2 bits is for types of instruction ($2^2 = 4$) and 4 bits for operations ($2^4 = 16$).

ISA format of ALU Instruction (Immediate) is:

Opcode (6 bit)		Register 1	Value	Unused
2 bits	4 bits	5 bits	16 bits	5 bits
(00) Types of instruction	Operations	Ra(00000-11111)	Value (0000000000000000-1111111111111111)	XXXXX

Since no of Registers is 32, 5-bits will be needed to address 32 registers ($2^5 = 32$). Since CPU is 16-bit, size of value will be 16-bit. Size of ISA is 6+5+16 = 27 bits. Assuming opcode of AND RA, 2 is 00000.

Example: ISA Design

ISA format of Jump Instruction is:

Opcode (6 bit)		Address	Unused
2 bits	4 bits	7 bits	19 bits
(01) Types of instruction	Operations	Value (0000000-1111111)	XXXXXXXXXXXXXXXXXXX

Since size of RAM is 128, it will take 7-bits to address all memory locations ($2^7 = 128$). Size of ISA is 6+7 = 13 bits. Assuming opcode of JMP LABEL is 010000.

ISA format of Memory Instruction (LOAD RA, [Address]) is:

Opcode (6 bit)		Register 1	Address	Unused
2 bits	4 bits	5 bits	7 bits	14 bits
(10) Types of instruction	Operations	RA (00000-11111)	Value (0000000-1111111)	XXXXXXXXXXXXXX

Size of ISA is 6+5+7 = 18 bits. Assuming opcode of LOAD RA, [Address] is 100000.

So, total size of ISA must be 27 bits. Extra bits in other instructions will be unused.

Example: ISA Design

ISA format of ALU Instruction (Immediate) is:

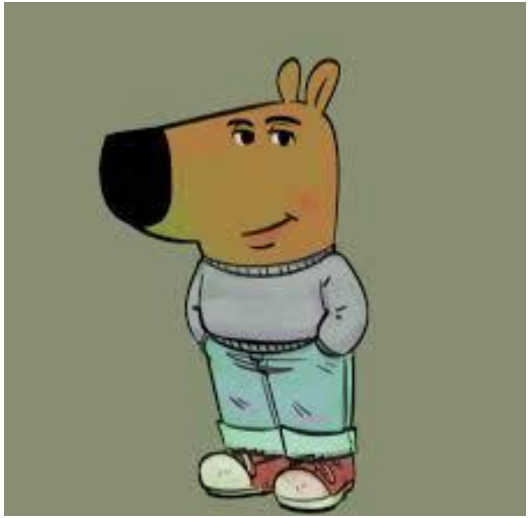
Opcode (6 bit)		Register 1	Value
2 bits	4 bits	5 bits	16 bits
(00) Types of instruction	Operations	Ra(00000-11111)	Value (0000000000000000-1111111111111111)

ISA format of Jump Instruction is:

Opcode (6 bit)		Address	Unused
2 bits	4 bits	7 bits	14 bits
(01) Types of instruction	Operations	Value (0000000-1111111)	X

ISA format of Memory Instruction (LOAD RA, [Address]) is:

Opcode (6 bit)		Register 1	Address	Unused
2 bits	4 bits	5 bits	7 bits	9 bits
(10) Types of instruction	Operations	RA (00000-11111)	Value (0000000-1111111)	X



Now I am a chill guy.
Thanks